

Communications Subsystem Hardware Reference

1. Overview

This document describes the **hardware setup of the ARGUS CubeSat communications subsystem**, including the spacecraft side and ground station side. It provides a clear reference for hardware integration, connectivity, and configuration. It also links to supporting documentation such as the *CubeSat Antenna Build & Tuning Guide*.

---

2. Spacecraft Communication Hardware

2.1 Antenna

- **Type:** Deployable tape-measure antenna
- **Frequency Band:** 433 MHz
- **Polarization:** Omnidirectional
- **Length & Width:** each leg about 17.3cm, width <6mm
- **Deployment Mechanism:** burn wire release, spring-loaded tape measure mechanism
- **Reference:** See *CubeSat Antenna Build & Tuning Document* for fabrication and tuning procedures.

2.2 Radio Module (LoRa)

- **Model:** Ebyte E22-400M30S
- **Frequency:** 433 MHz (dependent on licensing)
- **Spreading Factor (SF):** 7-12 depending on communication mode (see communication protocol documentation once determined)
- **Coding Rate:** 5

- **Bandwidth:** 125 kHz
- **Transmit Power:** 1W (30 dBm)
- **Interface:** SPI to mainboard RP2350
- **Notes:** link to circuit python driver ([insert link here])

2.3 RF Cabling & Connectors

- **Cable Type:** U.FL to U.FL coaxial cable
- **Cable Length:** 15 cm
- **Connector Types:** U.FL (radio side) → U.FL (antenna interface)
- **Integration Notes:** Ensure proper strain relief, cable routing, and RF shielding to minimize loss.

---

3. Ground Station Communication Hardware

3.1 Antenna

- **Type:** Eggbeater antenna
- **Frequency Band:** 433 MHz
- **Polarization:** Omnidirectional, circularly polarized
- **Mounting/Pointing Setup:** Fixed

3.2 RF Front-End

- **AS-3000 System Switch**
  - Purpose: RF switch for routing signals between Tx/Rx chains
  - Datasheet: SSB Electronic AS-3000
- **LNA 5000 Wide Band Preamp** (2 units)
  - Band: 50 – 5000 MHz
  - Purpose: Wideband low-noise amplifier for receive chain
  - Datasheet: SSB LNA 5000
- **Q5 Signal 7060 PAS Linear Amplifier (1W Input, 50W Output, 70cm)**
  - Purpose: Linear amplifier for transmission
  - Specs: ~50W at 70cm band, 1W input
  - Datasheet: Q5 Signal 7060PA

3.3 Cabling & Connectors

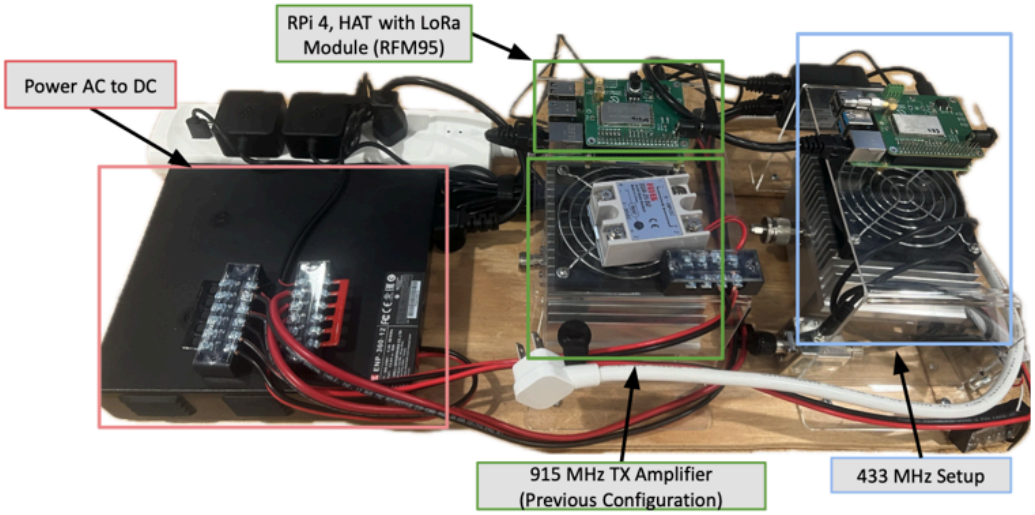
- **SMA Male-Male Adapter** (1x)
- **SMA Female-Female Adapter** (2x)
- **BNC–SMA Adapters**
- **Type N Male–SMA Male Adapter**
- **4-Pin DIN Connector** (for LNA switch)
- **Low-loss coaxial cables with SMA connectors** (various lengths as required)

3.4 Switching & Power

- **AC → DC Power Converter**
  - Purpose: Provides regulated DC supply for RF amplifiers, LNAs, relays, and Pi HAT board.
  - Input: *? Unsure, will update asap*
  - Output: *? Unsure, will update asap*
  - Notes: Ensure grounding and RF noise isolation.
- **Solid State Relays (SSR-40DA, 2x)**
  - Used for LNA switch control and other RF routing logic
- **Zip Ties (UV-rated)**
  - For securing cables and mechanical strain relief
- **Wooden Board / Mounting Plate**
  - Base plate for assembling RF chain and securing components

3.5 Control & Processing Unit

- **Raspberry Pi** (main control computer)
- **In-house designed Ground Station Pi Hat Board** (Techspark fabrication)
  - Provides GPIO interface for relays, RF switch, and LNA power routing
- **Software Stack:** [insert details: GNU Radio / custom Python LoRa scripts / logging system]



*** Add block diagram ***

---

4. Link Budget Parameters

UHF Uplink

	dBm	Linear	Units
Transmitter: Ground Station			
Frequency		433	MHz
Wavelength		0.69	m
TX Power	46.99	50	W
TX Antenna Gain	5.50	3.55	
Receiver: Spacecraft			
Total Losses	-148.01		
RX Antenna Gain	0.00	1.00	

RX Power	-94.52	3.53E-13	W
RX Sensitivity	-123.00	5.01E-16	W
Margin:	**28.48**		dB

UHF Downlink

	dBm	**Linear**	**Units**
Transmitter: Spacecraft			
Frequency		433	MHz
Wavelength		0.69	m
TX Power	30.79	1.20	W
TX Antenna Gain	0.00	1.00	
Receiver: Ground Station			
Total Losses	-148.01		
RX Antenna Gain	5.50	3.55	
RX Power	-111.71	6.74E-15	W
RX Sensitivity	-124.50	3.55E-16	W
Margin:	**12.79**		dB

* These calculations were made assuming a spreading factor of 7 and coding rate of 5. To experiment with different values, link budget spreadsheet calculations can be found here.

5. Testing & Validation

5.1 Antenna Testing

- VSWR and return loss measurement.
- Radiation pattern characterization in anechoic chamber and outdoor range.
- Deployment testing for tape-measure antenna reliability.

5.2 Radio & Interface Testing

- Verify SPI/UART communication with onboard computer.
- Validate power consumption against budget.
- Test LoRa transmission/reception with configurable SF, bandwidth, and coding rate.

5.3 End-to-End Link Testing

- Ground ↔ CubeSat bidirectional link verification.
- Ridge testing over **≥1 mile** line-of-sight with ground station.
- Evaluation of packet error rate (PER) under varying conditions.
- See Ridge Test Procedure documentation for reference

---

6. Open Questions & Future Work

- Finalize **LoRa module selection and configuration** (model, SF, bandwidth).
- Assess **antenna deployment reliability** and redundancy needs.
- Explore integration with **mission operations software** for command/telemetry automation.
- Add as needed